NYU TANDON SCHOOL OF ENGINEERING

ROB-GY 5103
Mechatronics

Project Proposal: Botanist Research Platform

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This project aims to develop an automated irrigation system to support plant health research, controlled by a Basic Stamp 2 (BS2) microcontroller. This "Botanist Research Platform" will utilize a soil humidity sensor to monitor moisture levels, triggering a pump to water the plant, whose humidity level will be based on user input. The system also integrates a light sensor to prevent irrigation during direct sunlight, reducing evaporation and optimizing water usage.

To further enhance functionality, the system includes a "smart water tank" that tracks water levels and usage over time with an ultrasonic sensor. A buzzer will alert the user when the water supply is low, ensuring consistent maintenance. Additionally, sensor data will be logged to allow researchers to analyze the plant water consumption throughout the experiment. This platform aims to be a low-cost, self-sustaining solution for botanical studies, giving researchers valuable insights while minimizing manual labor.

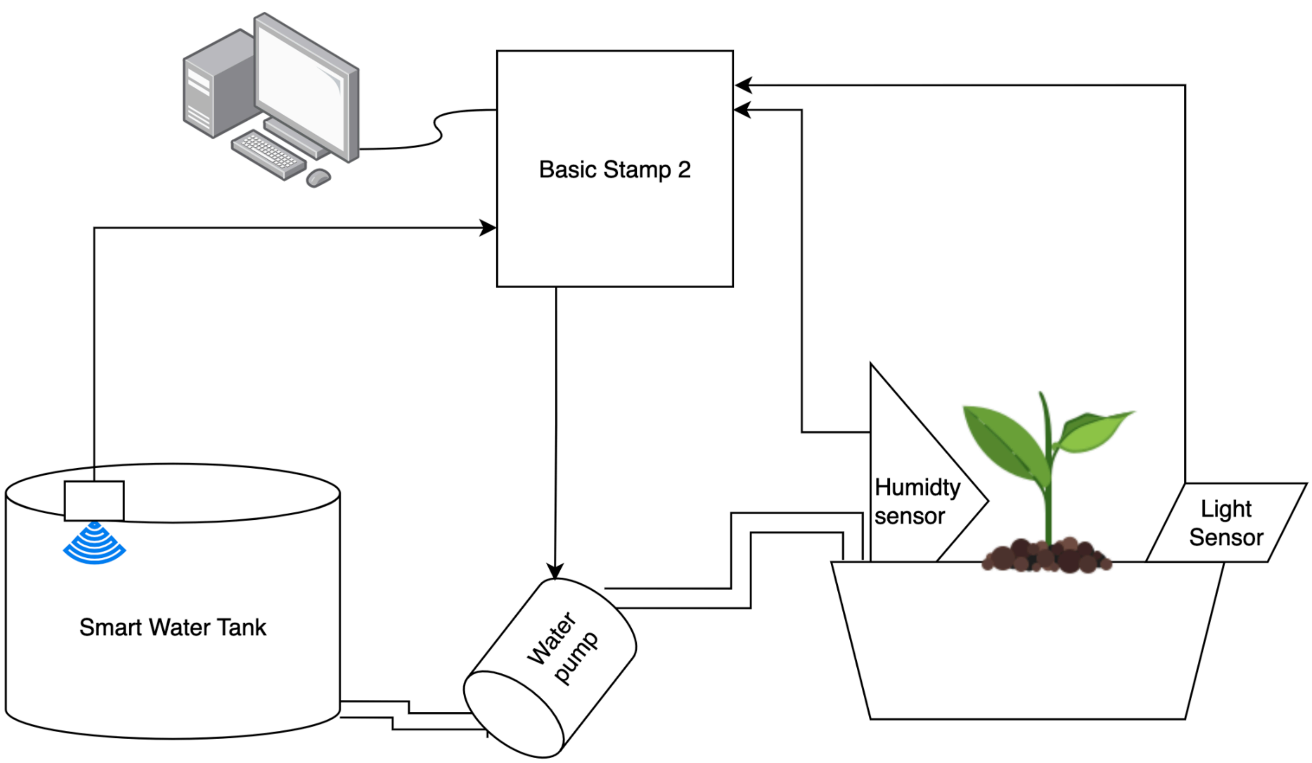

Figure 1: Botanist Research Platform Conceptual Diagram